G. Lejeune Dirichlet's Werke. II.

XXII.

Memorial Address on Carl Gustav Jacob Jacobi

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In undertaking to describe the scientific achievements of the greatest mathematician who, since Lagrange, has belonged to our body as a resident member, I feel vividly the difficulties of the task. I must set forth the full significance of the creations of a man who, with a strong hand, intervened in almost every region of a science which through two thousand years of labour has grown to immeasurable extent; who wherever he directed his creative mind brought important and often deeply hidden truths to light; and who, by introducing new fundamental ideas into science, raised mathematical speculation in more than one direction to a higher level. Only the conviction that a debt of gratitude is owed for such services rendered to science and to those who cultivate it can silence the reservations called forth in me by the consciousness of my inadequacy; for on whom could the fulfilment of this duty fall more than on me, who, like all my colleagues in the field, was so essentially advanced by Jacobi's scientific productions and who, besides, owed no less instruction to my many years of close intercourse with the great investigator.

Carl Gustav Jacob Jacobi was born on 10 December 1804 in Potsdam, where his father was a well-to-do merchant. His first instruction in the ancient languages and in the elements of mathematics was given by his maternal uncle, Mr. Lehmann, who had less to teach the active boy than to guide him. Under this intelligent guidance Jacobi made such rapid progress that, not yet twelve years old, he entered the second class of the Potsdam Gymnasium and after only half a year was admitted to the first. He remained there for a full four years, since he could not properly attend the university before completing his sixteenth year. The mathematical instruction, which was treated wholly as a matter of memory, could not suit the young senior pupil. His relation to the teacher was therefore for some time very disagreeable; in the end it improved, because the teacher had the good sense to let the unusual pupil go his own way and to permit him to occupy himself with Euler's Introductio while the other pupils recited elementary propositions that had been laboriously learned. How far Jacobi's intellectual development had already advanced is shown by the attempt he made at about this time to solve equations of the fifth degree, an attempt which he later mentioned in one of his papers.

[1] The corresponding note in the Academy report for 1852 reads on p. 433: "Hereupon Mr. Dirichlet delivered a memorial address on the deceased member of the Academy, the mathematician Jacobi." K.

Many of those who later acquired a great name first tested their powers on this problem; and it is indeed easy to understand what attraction this problem must have exercised on an awakening talent as long as its impossibility had not yet been proved. To the fame which so many fruitless efforts had given this investigation there was added the special circumstance that the problem, belonging to a domain which borders immediately on the elements, seemed accessible without a large store of preliminary knowledge.

At the university here Jacobi divided his time among philosophical, philological, and mathematical studies. As a participant in the exercises of the philological seminar he attracted the attention of our colleague Böckh, the head of that institute, who became very fond of the young man because of his keen and distinctive mind and distinguished him by special goodwill. He seems to have attended few mathematical lectures, since at that time they had too elementary a character at this university to advance Jacobi essentially; he was already familiar with some of the chief works of Euler and Lagrange. All the more zealously he surveyed the mathematical literature, seeking in particular to gain a general overview of the great scientific treasures contained in the academy collections.

Jacobi's nature was not satisfied with the mere gathering of information. He felt the need to become fully master of the things with which he occupied himself. After about two years of university study he recognized the necessity of making a decision and of renouncing either philology or mathematics. Since the decision he made had important consequences not only for him but also for the science to which from then on he devoted himself exclusively, one will gladly hear from him the reasons that determined his choice. He wrote on this subject to the uncle already mentioned: "Although for a time I occupied myself seriously with philology, I did at least succeed in gaining a glimpse into the inner splendour of ancient Hellenic life, so that I could not give up its further investigation without a struggle. But for the present I must give it up entirely. The immense colossus which the labours of Euler, Lagrange, and Laplace have called forth demands the most immense force and exertion of thought if one is to penetrate into its inner nature, and not merely rummage about its exterior. A drive that will not rest or pause compels one to become master of this structure, so that one need not fear at every moment to be crushed by it, until one stands above and can survey the whole work. Only then is it possible to work calmly and properly on the perfection of its individual parts, and to carry the great whole further according to one's powers, once one has grasped its spirit."

For his doctoral dissertation Jacobi chose a subject already often treated, the decomposition of algebraic fractions. In it he first proved remarkable formulae which Lagrange had given without proof in the memoirs of our Academy; he then passed to a new kind of decomposition, which unlike the one exclusively considered until then is not completely determinate; and he concluded the paper with investigations on the transformation of series, in which a new principle already becomes noticeable, one of which he made repeated use in later works.

Immediately after his promotion Jacobi habilitated at the university and gave a lecture on the theory of curved surfaces and curves in space. According to the testimony of one of his hearers at that time, his teaching talent must already have been highly developed when he first appeared, and he must have known how to treat his subject with great clarity and in a way very stimulating for his audience. The twenty-one-year-old lecturer also showed very early maturity of judgement in that, undisturbed by the discredit into which the method of the infinitely small had at that time fallen through a great authority, he followed precisely this method in his exposition and sought, with the best success, to convince his hearers that the suspected method differs from the strict method of the ancients only in its abbreviated form, but becomes indispensable precisely through this form in all more complex questions.

The attention Jacobi began to attract caused the highest educational authority to ask him to continue his teaching activity provisionally as a Privatdocent in Königsberg, where the professorship of mathematics, just then vacant, gave him greater prospects of promotion than did Berlin. On moving to Königsberg it was an important event for Jacobi to become personally acquainted with the great astronomer Bessel, and for the first time to see, close at hand and in a field so nearly related to his own, a genius. The daily sight of the fiery zeal of this extraordinary man exercised the mightiest influence even on Jacobi, though he had been accustomed from his earliest youth to demand the greatest efforts from himself; he often mentioned this influence later with gratitude.

It was a fortunate circumstance for Jacobi's literary career that its beginning coincided with the founding of the mathematical journal through whose publication our colleague Crelle acquired so great and lasting a merit, not only for the dissemination but also for the enlivening of the study of science. Jacobi, who belonged to the earliest collaborators of the journal, remained faithful to it until his death; and, apart from the two separate works Fundamenta nova theoriae functionum ellipticarum and Canon arithmeticus, almost all his other works first appeared in Crelle's Journal.

Jacobi's first papers already show him as a fully mature mathematician, whether, as in the essays on Gauss's new method for the approximate determination of integrals and on Pfaff's method for the integration of partial differential equations, he was considering known theories from a new point of view and essentially simplifying them, or whether he was treating problems not yet solved and arriving at new results. Among works of the latter kind, two should be especially mentioned here: a paper of a few pages in which he teaches a hitherto unknown fundamental property of the remarkable function first introduced into science by Legendre, which plays so great a role in all later general investigations on attraction; and another paper on cubic residues. The latter contains only theorems without proofs, but these theorems are of such a kind that they cannot be the result of induction and leave no doubt that Jacobi already then possessed new and fruitful principles in the scientific field that Gauss had opened to mathematical speculation a quarter of a century earlier, and that belongs as much to higher algebra as to the theory of numbers. This is also confirmed by a later publication in which he expressly mentions that he had already at that time communicated these principles to Gauss by letter.

Jacobi was then drawn away from the further pursuit of this subject by another work: his investigations on elliptic functions, which were soon to bring him so great a fame and to assign him a place among the foremost mathematicians of the age. The young mathematician, who had already tried his powers successfully in so many directions, seemed for a longer time to be not favoured by fortune in the theory of elliptic functions. One of his friends, who one day found him strikingly out of spirits, asked the reason for this mood and received the answer: "You see me just on the point of sending this book, Legendre's Exercices, back to the library; I have had decided bad luck with it. When I have otherwise studied an important work, it has always stirred me to thoughts of my own and something has always fallen to me from it. This time I have come away quite empty and have not been inspired by the slightest idea."

If his own thoughts were somewhat long in coming in this case, they appeared later all the more abundantly - so abundantly that, together with Abel's simultaneous thoughts, they produced an unexpected extension and the complete transformation of one of the most important branches of analysis. Since progress here proceeded at the same time from two different sides, it is necessary, beside Jacobi's investigations, to mention Abel's simultaneous works. Though independent of one another in origin, the discoveries of both later interlocked so closely that the exposition of one would be scarcely comprehensible without consideration of the other.

The theory of elliptic functions, with which the names of Abel and Jacobi are forever connected, does not in its beginnings reach back beyond the first half of the preceding century. An Italian mathematician of unusual acuteness, Count Fagnano from the Papal States, made the remarkable discovery that the integral which expresses the arc of the curve then much studied by mathematicians under the name of the lemniscate possesses properties similar to those of the simpler integral that represents a circular arc; and that, for example, between the limits of two integrals of this kind, one of which is equal to twice the value of the other, there is a simple algebraic connection. Thus an arc of the lemniscate, though a transcendent of higher kind, can nevertheless, like a circular arc, be doubled or halved by a geometrical construction[2]. Euler found, a few years later, the true source of this and of other similar properties in a theorem that belongs among the finest enrichments which science owes to that great investigator.

According to Euler's theorem, a certain integral, more general than that considered by Fagnano and in our present terminology called an elliptic integral of the first kind, depends on its limit in such a way that two such integrals with arbitrary limits can always be united into a third whose limit is a simple algebraic combination of the limits of the former, just as the sine of an arc composed of two parts can be formed algebraically from the sines of its parts. But the elliptic integral is more general than the one that expresses a circular arc. Brought to its simplest form, it depends not, as that one does, merely on its limit, but also on another magnitude contained in the function, the so-called modulus. Euler's theorem gave only relations among integrals with the same modulus. The first example of a connection among integrals that differ in their moduli was offered by a later transformation due to Landen and, in a somewhat different form, to Lagrange. Such a transformation, combined with Euler's theorem, leads in particular to a theorem which includes Fagnano's duplication as a special case.

The theory as Abel and Jacobi found it presented several highly enigmatic phenomena for whose clarification the principles then known did not suffice. Thus, to mention only one of these phenomena, it had been found that the degree of the equation by means of which the division of an elliptic integral into an arbitrary number of equal parts is effected is the square of the number of parts. Since even the division by two already leads to an equation of the fourth degree, the problem of division, notwithstanding Fagnano's discovery, seemed to belong to the class of questions that are algebraically insoluble. In the same way, the transformations that had become known through Landen and Lagrange appeared to stand isolated, and there was no consciousness before Abel and Jacobi of any general principle by which they might be connected.

[2]Fagnano's theorem is found in the *Giornale de' Letterati d'Italia*, tomo 29, Anno 1718. F.

The first fundamental idea by which the theory was transformed was the inversion of the elliptic integral. Earlier mathematicians had viewed the integral as a function of its limit. Abel and Jacobi recognized that the inverse function, the limit regarded as a function of the integral, must be made the proper object of study. This simple change of viewpoint opened the way to the true analogy with the circular functions. A second idea, common to Abel and Jacobi, was the introduction of imaginary quantities. Jacobi later often repeated that this introduction of the imaginary was the true key to the nature of the elliptic functions. Indeed, when one considers how widely Euler had already used imaginary quantities in the theory of the trigonometric functions, it would have been surprising had this thought escaped him entirely. Yet in the new domain it required a special insight. Abel and Jacobi, by introducing the imaginary into the functions obtained by inversion from the elliptic integral of the first kind - those which we now call elliptic functions - recognized that these functions share at once in the nature of the circular functions and in that of exponential quantities, and that while the former are periodic only for real and the latter only for imaginary values of the argument, elliptic functions unite both kinds of periodicity.

Placed by these fundamental ideas on new ground, Abel and Jacobi directed their investigations to two different regions of the theory. Abel's activity turned to the problems concerning the multiplication and division of elliptic integrals. By means of the principle of double periodicity he penetrated deeply into the nature of the roots of the equation on which division depends, and arrived at the wholly unexpected discovery that the general division of the elliptic integral with arbitrary limit can always be performed algebraically, that is, by mere extraction of roots, as soon as the special division of the so-called complete integrals is assumed already carried out. This special division seems possible only for special moduli, among which the simplest is the one corresponding to the lemniscate. Carrying out the solution for this case, he showed that the division of the whole lemniscate is completely analogous to the division of the circle and can be achieved by geometrical construction in precisely the same cases in which, according to Gauss's beautiful theory given twenty-five years earlier, the circle admits such a division.

A noteworthy historical circumstance attaches to this last work of Abel. In the introduction to the last section of the Disquisitiones arithmeticae, which is devoted to the division of the circle, Gauss had remarked in passing that the same principle on which his cyclotomy rests is also applicable to the division of the lemniscate. In fact, the Gauss principle according to which the roots of the equation to be solved are to be brought into a cycle so that each depends on the preceding one in the same way lies essentially at the foundation of Abel's paper on the division of the lemniscate. But while, for the division of the circle, long-known properties of trigonometric functions sufficed to arrange the roots according to Gauss's principle, in the case of the lemniscate an insight into the nature of the roots was required for a similar arrangement, indeed even to recognize the possibility of such an arrangement; and only the principle of double periodicity could provide that insight. Gauss's earlier remark therefore became, through Abel's paper, an incontrovertible testimony that, far in advance of his time, he had already at the beginning of the century recognized the principle of the double period. This testimony, however, first became intelligible through Abel's later work and therefore detracts nothing from the right of Abel and Jacobi to this discovery.

Besides the results just mentioned relating to division, Abel's investigations had another, no less important consequence. Letting the multiplier become infinite in the formulae by which he had represented the elliptic functions of a multiple argument by the functions of the simple argument, he obtained remarkable expressions for the elliptic functions in the form of infinite series and also as quotients of infinite products - a discovery perhaps of even greater importance for analysis than Abel's demonstration of the algebraic solvability of the equations for division.

At the same time as Abel was carrying out these beautiful investigations, Jacobi was no less successfully occupied in another part of the same field. The substitution mentioned above, by which an elliptic integral is transformed into an integral of the same form, had until then been the only one of its kind. Legendre had, it is true, found a second transformation of elliptic integrals not long before Jacobi turned to this subject; but this second transformation, with which Legendre considered the subject closed, was not then known in Germany. It therefore required rare acuteness to infer from a visible ring the existence of an infinite chain, and equally great boldness to set oneself the task of recognizing the nature of that chain.

A fortunate induction, in which the subtle and wholly new idea played an essential role of considering transformation and multiplication from a common point of view and the latter as a special case of the former, led Jacobi to the conjecture that rational functions of every degree are capable of transforming an elliptic integral into an integral of the same form. This conjecture was at once confirmed, for it appeared that the number of arbitrary coefficients available in each degree sufficed to satisfy all conditions that had to be fulfilled if the transformed integral was to agree in form with the original one. Yet even if so simple a consideration could hardly leave any doubt concerning the possibility of the matter, a great step still had to be taken in order to recognize the inner analytical nature of the fractional expressions suited to transformation. What sort of difficulties had to be overcome here, and by what ingenious considerations Jacobi overcame them, cannot be set out here; nor am I permitted to enumerate all the important consequences that followed from the completely solved problem. I mention only the remarkable result of this investigation, that multiplication can always be composed of two transformations.

While Abel and Jacobi were thus perfecting the theory simultaneously in two different directions, it seemed as though fate wished to divide evenly between the young rivals the honour of the advance to be accomplished. For the way in which each soon afterwards carried further the other's invention left no doubt that either one, had the other not anticipated him in part of the work, would have accomplished the whole advance by himself. Jacobi had proceeded in his investigations from the assumption that in the transformation the original variable is rationally expressed by the new one. Abel treated the problem under the more general hypothesis that some algebraic equation holds between the two variables, and reached the result that the problem thus generalized can always be reduced to the case which Jacobi had treated so completely.

Jacobi intervened no less successfully in the theory of general division given by Abel. The manner in which Abel had solved the problem showed indeed that the roots are always algebraically expressible, but for their actual representation it required the formation of certain symmetric combinations of roots, which could be carried out only in each particular case. From a new principle, to be mentioned more closely in a moment, Jacobi derived the final expressions for the roots, valid for every degree and formed directly from the data of the problem. These expressions had, moreover, a greater simplicity of form than Abel's. When Jacobi published the result of this work in a brief note, he hoped to astonish Abel by the perfection of the solution of the division problem; but this hope remained unfulfilled. Abel had just died, scarcely twenty-seven years old, less than two years after the publication of his first works on elliptic functions. Death had set so early a limit to the brilliant career of this profound and comprehensive mind.

Jacobi's further investigations on elliptic transcendents, as well as the one just mentioned, arose from a thought which, because of its consequences, must perhaps be assigned the first place among his conceptions. It was the idea of introducing into analysis, as independent transcendents, the infinite products by whose quotients Abel had expressed the elliptic functions. Once he had succeeded in representing these products, which are all of the same nature and may be regarded as special cases of a single transcendent, in the form of series, he recognized a function that had already presented itself to French mathematicians in investigations of mathematical physics, but had been little noticed there and of which only one property had been observed. Jacobi subjected it to a deeply penetrating investigation, explored its analytical nature, and then introduced it into the theory of integrals of the second and third kind. This led not only to the recognition of the inner connection among already known but isolated properties of these integrals, but also to the important discovery that the integrals of the third kind, which depend on three elements, can be expressed by means of the new transcendent, which contains only two.

In Jacobi's later exposition of the whole theory, as he was accustomed to give it in his lectures, the consideration of this function formed the starting point. The whole doctrine thereby gained not only a surprising degree of simplicity and transparency; this reversed order is also noteworthy because it became the model for other investigations to be mentioned later. If one considers that the new function now rules the whole domain of elliptic transcendents, that Jacobi derived important theorems of higher arithmetic from its properties, and that it plays an essential role in many applications - among which I mention here only the representation by means of this transcendent of the rotational motion that forms one of Jacobi's last and finest works - one must assign to this function the next place after the elementary transcendents long received into science. Strikingly, so important a function has as yet no other name than the transcendent Theta, after the accidental symbol with which it first appears in Jacobi; mathematicians would only fulfil a duty of gratitude if they agreed to attach Jacobi's name to it, in order to honour the memory of the man among whose finest discoveries belongs the first recognition of the inner nature and high significance of this transcendent.

Abel's works mentioned above are not the only first-rank achievement of this outstanding mathematician; they are not even the most important of his achievements. His greatest discovery is contained in a theorem that bears his name and wholly bears the stamp of his extraordinary spirit, whose characteristic feature was to treat the questions of science in the most comprehensive generality. The Euler theorem already described - I speak here of the principle itself, not of the consequences drawn from it, which were daily extended - then formed, in the domain to which it belongs, the boundary of science, beyond which only conjecture could reach. Abel's theorem suddenly opened an immeasurable region to inquiry. Legendre called Abel's theorem a monument more lasting than bronze, and Jacobi designated the same theorem, as it appears in simple generality, as the greatest discovery of the century.

In its original form the theorem asserted that a sum of any number of elliptic integrals can always be reduced to a certain fixed number of integrals of the same kind. Abel's thought went far beyond elliptic integrals and applied to the integrals of arbitrary algebraic functions. It is not possible here to indicate even in outline the whole range of this theorem and the problems to which it gives rise. This work has already begun, and Jacobi himself had the greatest share in it. Since a problem to be solved by Abel's theorem leads to the inversion of several integrals at once, and since this inversion would be impossible by the old means, one needed again a guiding function analogous to the elliptic theta-function. Jacobi indicated the way. Later, taking his treatment of elliptic functions as their model, two younger mathematicians of unusual talent, Goepel and Rosenhain, founded their beautiful works on the consideration of infinite series and thereby carried into the new theory the same kind of illuminating principle that Jacobi had introduced into the old one.

Although in the preceding account of Jacobi's discoveries in the field of elliptic and Abelian transcendents I have had to confine myself to the most essential points, this account has already grown to a length that forces me to touch only briefly on his remaining achievements. Yet even these brief indications will show how widely his mind ranged. Already above I spoke of his investigations on cyclotomy and on the form that Gauss first gave to the solution of binomial equations. In some of these results he met with the great mathematician Cauchy, who was occupied at the same time with similar researches and mentioned this circumstance when he published extracts from his works during Jacobi's first stay in Paris.

From a beautiful theorem derived from cyclotomy, also found by Cauchy, according to which all primes that, on division by a given prime or by four times it, leave the remainder one are represented, when raised to a certain power whose exponent depends only on the latter prime, by the so-called principal quadratic form having the negative of the given prime as determinant, Jacobi drew the conjecture that this exponent must agree with the number of distinct quadratic forms corresponding to that determinant. Since this conjecture was confirmed in all numerical examples, he did not hesitate to publish the observation in a brief note. I believe I ought to mention here, following Jacobi's oral communication, the hitherto unknown origin of this result as a remarkable example of acute induction, although the rigorous proof of it does not seem capable of being based on cyclotomy, but appears to require essentially different principles drawn from integral calculus and the theory of series, principles that were introduced into science only later.

Gauss's second memoir on biquadratic residues, published in 1832, made an epoch by the profound idea of treating complex integers in higher arithmetic just as real ones are treated, and by the reciprocity law established there, which holds in the theory of biquadratic residues between two complex primes. This memoir gave Jacobi occasion to resume his earlier investigations, and he succeeded in deriving with great simplicity from cyclotomy the beautiful theorem of Gauss just mentioned and a similar theorem relating to cubic residues. Although Jacobi completely wrote down these investigations and others connected with them, which I cannot even indicate, in the years 1836 to 1839, he never came to publish them in print. His delay arose from the wish to give some of his results greater extension, for which, occupied by so many other works, he did not find the necessary leisure. A part of his researches, and especially the proofs of the reciprocity theorems already mentioned, have nevertheless become known to some German mathematicians through transcripts of the lectures which he gave in Königsberg in the winter of 1836-37 on cyclotomy and its application to number theory.

Jacobi discovered another highly productive source for higher arithmetic in the theory of elliptic functions, from which he derived beautiful theorems on the number of decompositions of a number into two, four, six, and eight squares, and others concerning numbers that are contained simultaneously in several quadratic forms. These important enrichments of science are a fruit of the above-mentioned introduction of the Jacobi function into the theory of elliptic transcendents.

Jacobi repeatedly occupied himself with the reduction and evaluation of double and multiple integrals. I mention here especially the simple method by which he reduced the determination of the surface area of a triaxial ellipsoid to elliptic integrals of the first and second kind, a reduction which Legendre, to whom it belongs among his finest achievements, had succeeded in obtaining only with the aid of very hidden properties of integrals of the third kind. In another paper belonging here Jacobi extended Euler's addition theorem to double integrals, and soon afterwards observed how Abel's theorem too was capable of a similar extension.

Only a part of Jacobi's works on this chapter of integral calculus was published. A large memoir on the attraction of ellipsoids, although almost completed for a long time, remained unprinted and became known only through a few occasional notes. When he was occupied with this problem he also came upon the beautiful theorem found by Poisson at about the same time, according to which the attraction exerted on a point in exterior space by an infinitely thin shell bounded by two concentric, similar, and similarly situated ellipsoidal surfaces can be represented without integral signs. Jacobi never publicly mentioned this circumstance, although he could have appealed to the testimony of several mathematicians to whom he had communicated the theorem before the first notice of Poisson's paper appeared.

Connected with these investigations is another work of Jacobi's that must not be passed over here because of its surprising result. Maclaurin, as is well known, first showed that a homogeneous liquid mass can, while preserving its external shape, rotate uniformly about a fixed axis if that shape is that of an ellipsoid of revolution. This beautiful result was later completed by d'Alembert and Laplace through the demonstration that to each value of the angular velocity, if it lies below a certain limit, there correspond two and only two such ellipsoids. Lagrange seems first to have thought of the possibility that a triaxial ellipsoid too might satisfy the conditions of permanence; at least, in his analytical mechanics, when treating this question, he starts from the more general assumption of three unequal axes. But without carrying the investigation through he believes he may conclude from two apparently insufficient equations that the equilibrium is impossible in this more general case. A similar hasty conclusion by an author of a well-known textbook first aroused Jacobi's suspicion. On closer consideration of those two equations he soon found, to his own and surely to every mathematician's great surprise, that a triaxial ellipsoid also satisfies the conditions of equilibrium. Thus a new branch was grafted on the old theory of rotating fluid masses.

The occasion given to Jacobi in his investigations on the attraction of ellipsoids by Poisson's theorem led him to other researches in geometry. Among them are his works on surfaces of the second degree and on the geodesic lines of an ellipsoid. The latter problem is connected with the calculus of variations and with analytical mechanics; Jacobi recognized that the path of a point moving freely on an ellipsoid under no forces is represented by geodesics, and he thereby brought into contact the geometry of surfaces, mechanics, and the theory of elliptic functions. Similar connections guided him in his treatment of Hamilton's great discovery on the integration of the equations of dynamics. It was characteristic of Jacobi that he did not rest content with using a result, but sought the simple thought lying behind it and then brought the whole matter into a form in which the calculation became transparent.

To Jacobi's most important investigations also belong those on Pfaff's problem, that is, on the integration of a differential expression with several variables. This problem had remained wholly fruitless until Jacobi freed it from an unnecessary complication and showed that, instead of the sequence of systems whose successive integration Pfaff had demanded, the treatment of the first alone makes all the rest superfluous; the first step of the older method therefore already leads completely to the goal. The improvement which Jacobi made in the calculus of variations has a similar character. For the existence of a maximum or a minimum the vanishing of the first variation is necessary, but this condition alone is not sufficient; the nature of the second variation decides whether a maximum, a minimum, or neither occurs. According to the theory as Jacobi found it, after the integrations demanded by the vanishing of the first variation, new integrations had to be performed in order to discuss the second variation. Jacobi showed that the former involve the latter, so that here too the complete solution of the problem is already given with the completion of the first step.

If it is the ever more prominent tendency of modern analysis to set thoughts in the place of calculation, there nevertheless remain certain domains in which calculation retains its rights. Jacobi, who advanced that tendency so essentially, also achieved admirable things in these domains by virtue of his mastery of technique. To this class belong his papers on the transformation of homogeneous functions of the second degree, on elimination, on the simultaneous values satisfying a number of algebraic equations, on the inversion of series, and on the theory of determinants. In the last-named chapter we owe to him a developed theory of the expressions he called functional determinants. Pursuing far the analogy of these expressions with differential quotients, he arrived at a general principle which he called the principle of the last multiplier, and which, in almost all integration problems occurring in applications, gives the means of carrying out the last integration by specifying a priori the integrating factor required for it.

The influence that Jacobi exercised on the progress of science would appear only incompletely if I did not mention his activity as a public teacher. It was not his task to transmit anew what was finished and handed down. His lectures moved entirely outside the realm of textbooks and embraced only those parts of science in which he himself had appeared as a creator; and with him that meant that they offered the richest abundance of variety. His lectures were not distinguished by the sort of clarity that may also fall to intellectual poverty, but by a clarity of a higher kind. Above all he sought to set forth the guiding ideas that underlie each theory. Removing everything that bore the appearance of artificiality, he developed the solution of problems so naturally before his hearers that they could form the hope of being able to create something similar. As he knew how to treat the most difficult subjects, he could rightly encourage his hearers with the assurance that in his lectures they would have to appropriate only quite simple thoughts.

The success of so unusual a mode of teaching, as I have just described it and as only a creative mind can command, was truly extraordinary. If now in Germany knowledge of the methods of analysis is spread to a degree unknown in any earlier time, and if numerous younger mathematicians extend and enrich science in every direction, Jacobi has had the most essential share in so gratifying a phenomenon.

Almost all were his pupils; rarely did a budding talent escape his attention; and no one, once he had recognized it, lacked his promoting counsel and his encouraging participation.

I have just tried to present Jacobi as inventor and in his activity as teacher. If I now venture to describe how he appeared outside the scientific sphere to those who stand far from the mathematical sciences, then I must indicate as the basic trait of his nature that he lived wholly in the world of thoughts, and that in him thinking had become, not something that even most significant people require a special effort to undertake, but a habitual state and as it were a second nature. When anything in life or in science had once attracted his attention, he did not rest until he had worked it up into thoughts of his own; and with this uninterrupted intellectual activity there was united in him so rare a memory that he could at once bring before himself and command everything with which he had once occupied himself.

The inexhaustible store of knowledge and of his own thoughts that stood at Jacobi's disposal at every moment, a rare intellectual mobility by which he knew how to adapt himself to every age and every capacity, and a characteristically humorous mode of expression that sharply designated things, gave the great mathematician an unusual significance also in social intercourse. This was heightened by his readiness to treat scientific questions extempore. This readiness sprang from the innermost essence of his nature, which found its true satisfaction in overcoming difficulties; and for him there was therefore a quite special attraction in making scientific results intelligible by simple considerations even to those who lacked the preliminary knowledge that seemed indispensable for them. Only he had to be convinced, in order to make such an attempt, that those with whom he spoke had a real interest in the matter.

Where, on the other hand, he thought he noticed thoughtless curiosity, or heard definite opinions pronounced with self-satisfaction by people who had never imposed on themselves the hard work of thinking for themselves, his patience left him, and he usually brought the conversation to an end by an ironic and often sharply dismissive remark. He was often reproached for having shown himself too conscious of his intellectual power on such occasions. But those who judged him so might perhaps have changed their opinion had they known the price for which he had acquired the right to such a consciousness. A letter from the year 1824, from a time when Jacobi was still completely unknown and therefore could have had no interest in describing his intellectual struggles in exaggerated colours, contains the following passage, which I communicate here literally as a remarkable contribution to the characterization of the extraordinary man. Jacobi had then just turned twenty and had for about a year been occupied exclusively with mathematical studies.

"It is hard work that I have done, and hard work in which I am engaged. It is not diligence and memory that lead here to the goal; here they are the most subordinate servants of pure thought in motion. But stubborn, brain-rending reflection demands more strength than the most enduring diligence. If, therefore, through constant exercise of this reflection I have gained some power in it, one should not think that it has become easy for me through some happy natural gift. Hard, hard work I have to endure, and the anxiety of reflection has often shaken my health mightily. The consciousness, to be sure, of the power gained gives the finest reward for the work, and at the same time the encouragement to continue and not grow slack. Thoughtless people, to whom that work and that consciousness are therefore wholly foreign, seek to spoil this consolation, which alone can prevent one from letting courage sink on the difficult path, by making hateful under the name of conceit or arrogance the consciousness of being one's own free self - for only in the movement of thought is the human being free and with himself. Everyone who carries within himself the idea of a science cannot do otherwise than estimate things according to the way the human mind reveals itself in them. By this great measure, much must therefore appear insignificant to him which may seem quite praiseworthy to others. Thus people have often accused me of arrogance too; or, as they have most beautifully praised me when they meant to utter blame, I am proud toward everything lower and humble only toward the higher. But that infinite measure which one applies to the world within and without prevents all overestimation of oneself, because one always has before one's eyes the infinite goal and one's limited power. In that pride and that humility I shall always strive to persist, indeed to become ever prouder and ever humbler."

That it was no mere phrase in Jacobi when he said that he estimated things according to the manner in which the human mind reveals itself in them, and that he really treated everything that did not touch the world of thoughts, if not with indifference then with equanimity, he showed in the most difficult situations of his life. This truly philosophical equanimity was revealed most admirably when the misfortune befell him of losing the entire fortune inherited from his father - a loss that might have been all the more painful to him because he had been married for ten years and had to care for a numerous family. Whoever saw him then, when he had hurried to assist with counsel and deed his mother, who had suffered a similar loss, could perceive not the slightest change in his mood. He spoke with the same interest as always about scientific things and complained only that the unexpected journey had torn him away from an investigation that was then occupying him vividly.

I have already mentioned how Jacobi's cult of thought made itself known in his recognition of Abel's great discovery. He showed a similar sense for everything intellectually significant, and the saying of an ancient writer does not apply to him, that people really admire only what they believe they themselves could accomplish. His recognition embraced the whole intellectual domain; and in his science Jacobi's joy over another's invention was all the more lively the more that invention differed in stamp from his own creations. It was a natural impulse in him in such a case to strengthen the expression of his approval by the admission that he himself would never have had that thought.

It remains only to complete, in a few words, what I mentioned above concerning Jacobi's external circumstances. When he began to make known his investigations on elliptic functions, he was still a Privatdocent. The admiration that his discoveries aroused among all those competent to judge such matters had the consequence that he was at once promoted to extraordinary professor and soon afterwards to ordinary professor.

In speaking of the reception that Abel's and Jacobi's discoveries - for here the names of both are inseparable - found among all specialists, I cannot refrain from mentioning by name the man who, through his many years of research, was especially called to appreciate the unexpected advance in its full significance. Legendre, who had so often accused his contemporaries of lack of interest and not long before had expressed his regret that his favourite science, abandoned by all others, had by him alone after forty years of labour, as he believed, been brought to completion, greeted Abel's and Jacobi's discoveries, which carried the theory far beyond the limits that seemed to him to be set by the nature of the subject itself, with such warm and indeed enthusiastic recognition that it is difficult to say whom such recognition honoured more: the young mathematicians to whom it came at the entrance of their career, or the noble old master who, almost at the goal, showed himself capable of such warmth of feeling.

A no less honourable distinction followed soon afterwards when the Paris Academy, although it had opened no prize competition on the theory of elliptic functions, recognized Abel's and Jacobi's works as the most important discovery of the time by awarding one of its great mathematical prizes and dividing it between Jacobi and Abel's heirs. I must restrict myself here to mentioning the proofs of recognition that marked Jacobi's entrance into the scientific career. The limits set for me do not allow me to enumerate all the distinctions that were later bestowed on him in so rich a measure, and whose mention ought not to be absent from a detailed biography.

Soon after Jacobi published in 1829 his Fundamenta nova theoriae functionum ellipticarum, which contains only a part of his investigations on this subject, he made his first extended journey abroad. He took the route through Göttingen in order to make Gauss's personal acquaintance, and then turned to Paris, where he stayed several months and where at that time, besides Legendre - with whom he had for some time been in close correspondence and for whom he always preserved great reverence - Fourier, Poisson, and other outstanding mathematicians who survived Jacobi were gathered.

A second journey abroad was undertaken by Jacobi, who since 1831 had been married to a woman of outstanding intellectual formation, only again in 1842, in the company of his wife. The occasion for this journey was so honourable for him that I cannot leave it unmentioned. To the enlightened statesman who then stood at the head of the administration in the province of Prussia, it seemed desirable in the interest of science that Bessel and Jacobi should once accept the invitation often addressed to them to take part in the learned assembly held yearly in England. He therefore submitted to the king the request that the costs of such a journey be granted; His Majesty deigned with royal munificence to comply with that request.

Soon after Jacobi's return from this journey, symptoms of an unfortunately incurable illness appeared. For a long time he hovered in the greatest danger. When this danger had finally for the moment been removed, his physicians declared that, to strengthen him, a longer stay in a southern climate was necessary. This medical opinion placed Jacobi in no small embarrassment, but the embarrassment was not of long duration. As soon as the situation became known through our colleague Alexander von Humboldt, whose weighty mediation is never absent where the honour of science and the welfare of its representatives are concerned, to His Majesty the King, a new act of royal generosity assigned a considerable sum for a journey to Italy.

The mild climate of Rome, where Jacobi spent the winter, had so beneficial an effect on him that those who saw him there, far from recognizing a convalescent in him, had to marvel at his truly extraordinary activity. During the five months of his stay there he not only wrote, besides several smaller essays that appeared in a scientific journal in Rome itself, an important and very extensive memoir intended for Crelle's Journal; he also undertook the comparison of the manuscripts of Diophantus preserved in the Vatican, with whom he had for some time occupied himself assiduously.

After returning to his fatherland he was transferred from Königsberg to Berlin, where the at least relatively milder climate seemed to threaten his health less. Without belonging here to the university, he had only the obligation to give lectures insofar as this would be compatible with the care that his state of health so greatly required. His literary activity during his residence here hardly fell behind that of the best Königsberg period, as is shown by the papers written here in about six years, which fill two substantial quarto volumes.

At the beginning of the year 1851 he had an attack of influenza to endure. Since, however, he recovered quickly and began again to work with great zeal, his friends might allow themselves the hope that he would long be preserved to them and to science, when on 11 February he suddenly fell ill again. His condition at once aroused the greatest concern; and when after several days it was recognized that he had been seized by smallpox, which on the ground undermined by his old illness showed the most malignant character, every hope vanished. On 18 February, at eleven o'clock in the evening, eight days after falling ill, he succumbed without struggle.

Jacobi's scientific career comprises exactly a quarter century, thus a far shorter period than that of most earlier mathematicians of the first rank, and scarcely half the time over which Euler's activity extended. With Euler he has the greatest similarity, not only through versatility and fertility, but also in that all the resources of science were always present to him and at his command at every moment.

Death, which so early and so suddenly took him away from work in possession of his full power, did not grant to science the great enrichments that it could still expect from Jacobi's never tiring activity. In saying this I do not speak merely on the assumption that in such a mind creative power could be extinguished only together with physical life; I also have before my eyes a series of almost completed works to which he himself could in a short time - perhaps during printing, as he liked so much to do in his last years - have put the finishing touch, and which must now be brought to light by his friends as fragments in imperfect form.

Even during his illness, scarcely four days before his death, he lamented the misfortune that had ruled over many of his larger works, interrupted by illness or domestic calamity. "When later," he added sadly, "I returned to work, I preferred to begin something new rather than take up again investigations that awakened such sad memories in me. But I see that I must no longer delay giving to the public those older works to which I have devoted so large a part of my best strength, if they are still to intervene fruitfully in the course of science. Fortunately, only a very short time is needed for this, which, I hope, will not be denied me."